In Calvin cycle the inputs are CO_2 from the atmosphere and the ATP and $NADPH_2$ produced from light reaction.

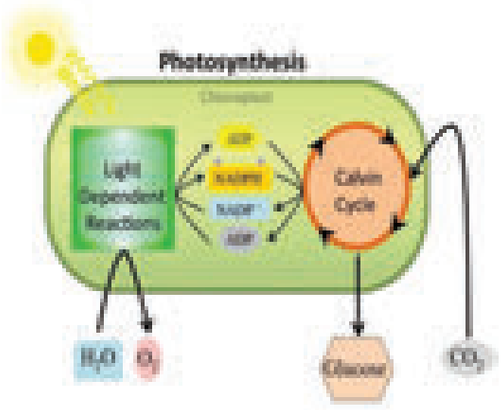


Figure 12.9 Overview of Hill and Calvin cycle



Melvin Calvin, an American biochemist, discovered chemical pathway for photosynthesis. The cycle is named as Calvin cycle. He was awarded with Nobel Prize in the year 1961 for his discovery.

12.9.8 Factors Affecting Photosynthesis

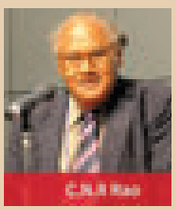
a) Internal Factors:

i) Pigments ii) Leaf age iii) Accumulation of carbohydrates iv) Hormones

b) External Factors:

i) Light ii) Carbon dioxide iii) Temperature iv) Water v) Mineral elements

Info bit



Artificial photosynthesis is a method for producing renewable energy by the use of sunlight. Indian scientist C.N.R. Rao who was conferred the Bharat Ratna (2013) is also working on similar technology of artificial photosynthesis to produce - Hydrogen fuel (renewable energy).

12.10 Mitochondria

Mitochondria are filamentous or granular cytoplasmic organelles present in cells. The mitochondria were first discovered by Kolliker in 1857 as granular structures in striated muscles. Mitochondria (singular: mitochondrion) are organelles within eukaryotic cells that produce adenosine triphosphate (ATP) which form the energy currency of the cell, for this reason, the mitochondria is referred to as the "**Power house of the cell**". Mitochondria vary in size from 0.5 µm to 2.0 µm. Mitochondria contain 60-70% protein, 25-30% lipids, 5-7% RNA and small amount of DNA and minerals.

12.10.1 Structure of Mitochondria

Mitochondrial Membranes: It consists two membranes called inner and outer membrane. Each membrane is 60-70 Å thick. Outer mitochondrial membrane is smooth and freely permeable to most small molecules. It contains enzymes, proteins and lipids. It has **porin molecules** (proteins) which form channels for passage of molecules through it.

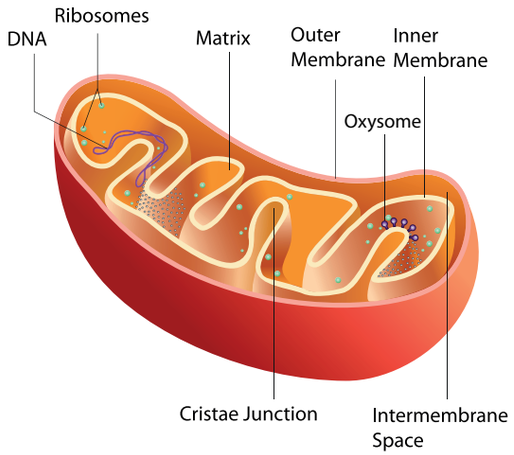


Figure 12.10 Structure of Mitochondria

Inner mitochondrial membrane is semi permeable membrane and regulates the passage of materials into and out of the mitochondria. It is rich in enzymes and carrier proteins. It consists of 80% proteins and lipids.

10th_Science_Unit-12.indd 182 11-01-2020 15:36:13